

ENHANCING SAFETY IN AUTONOMOUS NAVIGATION: WEATHER-AWARE TRAJECTORY PLANNING

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Abstract— In a context where autonomous navigation aims to enhance safety and comfort, this work introduces an adaptive weather routing algorithm for autonomous ships. Unlike traditional ships, where the final decision on the route is always entrusted to the captain, despite possible support from weather routing algorithms, on autonomous ships, it is necessary to automate the trajectory choice to protect cargo, structure, and, if present, crew engaged in activities, for example, research. The core of the study is the development of an efficient adaptive weather routing model that optimizes the ship's response in adverse marine conditions, implementing two approaches: the Dijkstra algorithm and the genetic algorithm. The comparison between these methods aims to determine which is more effective in adapting to such challenges, with the final goal of ensuring safer, more comfortable journeys with lower risks for cargo and ship.

Keywords — *autonomous navigation; adaptive weather routing; seakeeping performance; route optimization; Dijkstra algorithm; genetic algorithm*

I. INTRODUCTION

Autonomous navigation, a rapidly advancing field at the intersection of robotics, artificial intelligence, and maritime engineering, holds immense promise for revolutionizing transportation across various domains. By using cutting-edge technologies such as sensors, machine learning algorithms, and sophisticated control systems, autonomous navigation systems enable vehicles to operate and make decisions independently, without direct human intervention. From self-driving cars on roads to unmanned aerial vehicles in the skies, and autonomous vessels navigating the seas, these systems are poised to redefine how we travel and transport goods.

Autonomous navigation is currently experiencing significant advancements aimed at improving comfort and ensuring safety during navigation, with a specific focus on preserving structural integrity and cargo security, especially under adverse weather conditions. This study presents an innovative approach to trajectory planning for unmanned vessels, with the primary objective of enhancing overall safety and on-board well-being, should the vessel under study be manned, for example for research activities.

At the heart of this research lies an advanced route planning system that integrates traditional weather routing metrics with seakeeping parameters tailored to optimize comfort levels, mitigate the risk of motion sickness, and ensure navigation safety through precise vessel motion control strategies. This approach aims to establish a harmonious

balance between navigational efficiency and crew welfare, thereby enhancing the overall maritime transport experience.

Although existing literature on weather routing primarily focuses on route modifications to optimize fuel consumption, the impact of weather routing on fuel savings is considered only partially significant, as highlighted in previous studies [1,2]. Therefore, this paper emphasizes the influence of seakeeping performance in weather routing, which can be utilized based on the performance indices considered for various purposes, such as navigation support and a decision-making tool for captains and crew [1,2], or to enhance specific operations like the takeoff and landing of UAVs on a mother ship, as well as ensuring safer navigation for autonomous vessels.

The proposed method relies on an innovative adaptive weather routing model crafted to identify the optimal route that enhances ship performance in varying sea conditions. This model is designed to prioritize specific seakeeping criteria, ensuring the selection of routes that best optimize vessel capabilities in dynamic maritime environments. In this context, the application of two distinct optimization approaches is explored: the Dijkstra algorithm and the genetic algorithm. This comparison aims to evaluate the effectiveness of the two models in adapting to the dynamic and complex challenges of adverse marine conditions.

The algorithm is applied to an autonomous ship, assumed as a reference vessel, to investigate the effect of the selected seakeeping parameters and vessel speed on the best route selection. Hence, the main aim of the current research is to develop an adaptive weather routing model aiming to: (i) enhance the vessel's seakeeping capabilities in varying sea conditions, (ii) improve comfort and safety in autonomous navigation, especially in adverse weather, and (iii) carry out a sensitivity analysis to investigate the effect of each seakeeping parameter on the optimum route selection.

II. LITERATURE REVIEW

The weather routing problem was addressed by many authors and different approaches were embodied in the past. Building on the foundational work of James [3], Zoppoli [4], and Papadakis and Perakis [5], the initial generation of weather routing strategies primarily aimed to minimize voyage duration and fuel consumption. This early work often overlooked the impact of route selection on the ship's performance in rough seas. In recent years, however, the scope of voyage optimization has broadened to include

considerations of ship seakeeping performance as well as fuel efficiency. Consequently, second-generation algorithms, which are general optimization tools applied to ship voyage optimization, have emerged. These can be categorized into dynamic programming, genetic algorithms, and pathfinding methods. Subsequent advancements, like those by Hagiwara and Spaans in 1987 [6], incorporated fuel consumption considerations and automated the method. Among the most widely used techniques are multi-objective genetic algorithms [7–9], which address discretized nonlinear optimization problems stochastically. These algorithms base their solutions on the ship's route and velocity profile, represented by a set of parametric curves. While most existing weather route optimization methods aim to minimize fuel oil consumption, Pennino et al. [1–2] presented a new adaptive weather routing model based on Dijkstra's shortest path algorithm, focusing on maximizing the ship's performance during navigation. Szlapczynska and Smierzchalski [10] introduced a multi-criteria evolutionary weather-routing algorithm, employing an iterative population development process to yield a Pareto-optimal set of solutions. Similarly, Weintrit et al. [11] introduced a three-dimensional (3D) dynamic programming approach aimed at optimizing the ship's speed and heading angle during route planning. Pathfinding methods are another significant approach to weather routing optimization. Padhy et al. [12] utilized the Dijkstra algorithm to tackle the ship routing problem, while Veneti [13] enhanced this approach with a time-dependent bi-objective shortest path algorithm to optimize conflicting objectives, such as fuel consumption and expected risk along the route. An overview presented in [14] discusses the complementarity between weather routing and safe ship handling, highlighting their extensive implementation to achieve optimal and safe navigation conditions. Recent advancements in trajectory planning for autonomous vehicles, particularly Unmanned Aerial Vehicles (UAVs), have also explored similar optimization methods. For example, adaptive planning strategies based on bit-coded flight maneuvers [15], as well as integrated approaches employing Theta* and clothoid curves [16], have been investigated to enhance UAV performance in dynamic environments. These advances underscore the importance of applying robust optimization techniques to improve both safety and efficiency in autonomous navigation, especially under adverse weather conditions.

III. SOFTWARE DESCRIPTION

A. Code structure

The adaptive weather routing model operates through a defined sequence of steps, starting with the reception of crucial input data: weather information, vessel details (dimensions, Response Amplitude Operators, added resistance values), initial speed, and trip departure/arrival coordinates. Using this data, the model conducts the preliminary calculations, estimating speed reduction due to various sea states and then calculating the JONSWAP sea spectrum. Subsequently, the algorithm evaluates by identifying the RAOs (Response Amplitude Operators) of interest, the response spectra for roll, heave, and pitch motions, key ship motions influenced by waves. This spectrum evaluation determines how the vessel responds to different heading angles and speeds, crucial as weather and sea conditions directly affect ship behaviour.

Subsequently, the model computes different criteria along the ship's trajectory and determines the overall Seakeeping

Performance Index (SPI) function value based on these criteria. Using SPI as a metric, the model initiates an optimization process, exploring alternative routes through genetic and Dijkstra algorithms.

The goal is to find the optimal route by minimizing the 1's complement of the SPI, thus ensuring navigation safety. Once optimization concludes, the model provides the optimal route, considering expected weather, sea conditions, and vessel characteristics.

In enhancing weather routing for maritime navigation, the genetic algorithm offers an innovative method for determining optimal routes amidst variable weather conditions. The genetic algorithm, unlike Dijkstra's algorithm, doesn't rely on a fixed grid of points but evaluates waypoints defining the course between start and end points.

Dijkstra algorithm was investigated, known for its speed in yielding results. This algorithm constructs a minimum total length tree among nodes, identifying the best path by minimizing a designated cost function. A preliminary network of nodes spanning the intended route establishes constraints, with each node containing dynamic weather data. Arcs connecting adjacent grid points in the network incorporate a cost function to determine the optimal route. The algorithm is not applied in a static manner but is iteratively updated by adjusting the starting point based on the elapsed time and the distance already covered by the ship. The details on the construction of the node network and the use of the Dijkstra algorithm to optimize the route are reported in [1].

In this study, an optimization methodology using a genetic algorithm has also been applied. Genetic algorithms are inspired by natural selection and biological evolution, simulating the process of natural evolution by creating a population of potential solutions and evolving them over successive generations. Each potential solution, known as an individual, is assessed using a fitness function that measures its effectiveness or quality relative to the optimization goal. The fundamental aspects of genetic algorithms include selection, crossover, and mutation. Selection involves picking the best-performing individuals from the current generation to form the next generation. Crossover, or recombination, merges selected individuals to produce new offspring that inherit traits from both parents. Mutation introduces random alterations to individual solutions, fostering diversity and preventing early convergence to suboptimal solutions. For optimizing ship trajectories, the genetic algorithm was used to determine the most efficient route by considering various sea conditions and ship responses. Routes with Seakeeping Performance Index (SPI) values above a critical threshold are penalized, promoting the selection of safer and more effective paths. This iterative approach continues until an optimal or satisfactory route is identified, balancing the trade-offs between route efficiency and seakeeping performance.

The adaptive weather routing model integrates sophisticated algorithms and dynamic calculations to derive optimal routes, ensuring maximum safety and efficiency in maritime navigation.

B. Seakeeping responses

In this analysis, the seakeeping performance of the ship is assessed using six key criteria: (i) pitch motion amplitude, (ii) roll motion amplitude, (iii) relative vertical acceleration at the ship's forward perpendicular, (iv) probability of slamming

occurrence, (v) probability of green water on deck, and (vi) Motion Sickness Incidence (MSI). These criteria are evaluated based on the following equation:

$$SPI = \max \left\{ 0; \left(1 - \frac{RMS_{pitch}}{RMS_{pitch,lim}} \right) \cdot \left(1 - \frac{RMS_{roll}}{RMS_{roll,lim}} \right) \cdot \left(1 - \frac{RMS_{acc}}{RMS_{acc,lim}} \right) \cdot \left(1 - \frac{P_{sl}}{P_{sl,lim}} \right) \cdot \left(1 - \frac{P_{wd}}{P_{wd,lim}} \right) \cdot \left(1 - \frac{MSI}{MSI_{lim}} \right) \right\} \quad (1)$$

The RMS of the pitch motion amplitude is determined according to the following equation, based on the ship Response Amplitude Operator:

$$RMS_{pitch} = \sqrt{\int_0^\infty S_\zeta(\omega_e) Y_{pitch}^2(\omega_e) \left(\frac{\omega_e^2}{g} \right)^2 d\omega_e} \quad (2)$$

Where Y_{pitch} is the Response Amplitude Operator for the pitch motion. S_ζ is the wave spectrum, and ω_e is the encounter wave frequency.

The root mean square (RMS) of the roll motion amplitude is defined by the following equation.

$$RMS_{roll} = \sqrt{\int_0^\infty S_\zeta(\omega_e) Y_{roll}^2(\omega_e) \left(\frac{\omega_e^2}{g} \right)^2 d\omega_e} \quad (3)$$

Where Y_{roll} is the Response Amplitude Operator for the roll motion.

The RMS of the relative vertical acceleration is obtained by combining the heave and pitch motions at the ship forward perpendicular:

$$RMS_{acc} = \sqrt{\int_0^\infty \left| H_{heave}(\omega_e) - \bar{x}H_{pitch}(\omega_e) + \bar{y}H_{roll}(\omega_e) - e^{-ik(\bar{x}\cos\chi + \bar{y}\sin\chi)} \right|^2 \omega_e^4 S_\zeta(\omega_e) d\omega_e} \quad (4)$$

where H_{heave} , H_{pitch} , H_{roll} are the transfer function for heave, pitch and roll motions, respectively, $\bar{x}$ is the longitudinal distance (fwd+) of the ship forward perpendicular from its centre of mass, $\bar{y}$ is the transverse distance (stbd+), χ denotes the encounter angle between the ship heading and the prevailing sea state direction and $k = \omega^2/g$ is the wave number satisfying the deep-water condition.

As concerns the slamming occurrence, it is determined as follows:

$$p_{sl} = e^{-\left(\frac{V_{cr}^2}{2C_s^2 m_{2,r}} + \frac{d^2}{2C_s^2 m_{0,r}} \right)} \quad (5)$$

where $V_{cr} = 0.093$ is the threshold velocity, C_s is the swell up coefficient equal to 1 up to $F_N = 0.30$, and d is the ship draught at the forward perpendicular, $m_{0,r}$ and $m_{2,r}$ represent the 0 and second-order spectral moments of the ship relative motion as regards the sea surface. The assessment of the probability of green water on the forward deck structures adopts a comparable methodology:

$$p_{wd} = e^{-\frac{f_b^2}{2C_s^2 m_{0,r}}} \quad (6)$$

Where the notation f_b denotes the freeboard at the ship's forward perpendicular. Finally, the MSI is defined starting from the following formula:

$$MSI = 100\Phi(z_a)\Phi(z'_T) \quad (7)$$

Where Φ is the standard cumulative normal distribution, while z_a and z'_T are functions of exposure time to ship motions.

The limit values of the pitch amplitude and the MSI are based on the NATO STANAG, Standardization Agreement, 4154 criteria, while the remaining ones comply with the NORDFORSK 1987 criteria, as detailed in Table 1.

TABLE 1. CRITERIA LIMITS

Criteria	Limit value
RMS of pitch amplitude	1.5 degrees
RMS of roll amplitude	6 degrees
RMS of vertical acceleration at fp	0.275 g ($L \leq 100$ m) or 0.050 g ($L \geq 330$ m) *
Slamming probability	0.03 ($L \leq 100$ m) or 0.01 ($L \geq 330$ m) *
Green water on deck probability	0.05
MSI	20% after 4 hours

*at intermediate values, linear interpolation is applied.

IV. DATA INPUT: SHIP, ROUTE, AND METEOROLOGICAL DATA

The vessel selected for the simulation is a containership and its specifications are described as follows in the table 2:

TABLE 2. CONTAINERSHIP'S DIMENSIONS

Dimension	Value
Length between perpendiculars	175.0 m
Breadth	25.4 m
Design draught	9.5 m
Displacement	24539.0 t
Froude number	0.18
Initial velocity	14.5 kn
Pitch moment of inertia	43286796 tm^2
Waterplane area	3152 m^2
Transversal metacentric radius	7.52 m
Longitudinal metacentric radius	206 m
Block coefficient	0.572

It is important to note that the selection of a containership for this simulation implies that certain seakeeping indices may not be relevant. In particular, for container ships, it is crucial to ensure that there are no negative impacts on the ship's structures and to monitor movements and accelerations that could adversely affect the cargo. While criteria such as, for example, the MSI (Motion Sickness Incidence) may not be pertinent. Nonetheless, the developed model has been meticulously crafted to be adaptable across various vessel types, including scenarios like the present one where there might be no crew members actively navigating the ship. Instead, the vessel could be utilized for tasks such as research expeditions. In such contexts, ensuring the well-being of crew members is paramount. Furthermore, in the absence of a commanding officer to oversee navigation, it becomes imperative to incorporate an autonomous decision-making system which takes into account criteria such as MSI.

The route chosen as reference falls along the southern Indian Ocean. This could be interpreted as a small part of the famous European-Asian sea route, commonly known as the sea route to India or the Cape route, which is a sea route from the European coast of the Atlantic Ocean to the Asian coast of the Indian Ocean passing for the Cape of Good Hope and Cape Agulhas on the southern border of Africa.

During the "Age of Sailing," this route held significant importance, yet its relevance diminished somewhat following the opening of the Suez Canal in 1869. However, it remains significant for vessels unfit for the canal or in closed situations.

The selected journey has the starting coordinates (38°00' S, 33°00' E) and the destination coordinates (31°30' S, 53°00' E), as illustrated in the figure 1, which also depicts the great circle or orthodromic (black) and the loxodromes or rhumb lines (blue), measuring 1058 and 1064 nautical miles respectively.

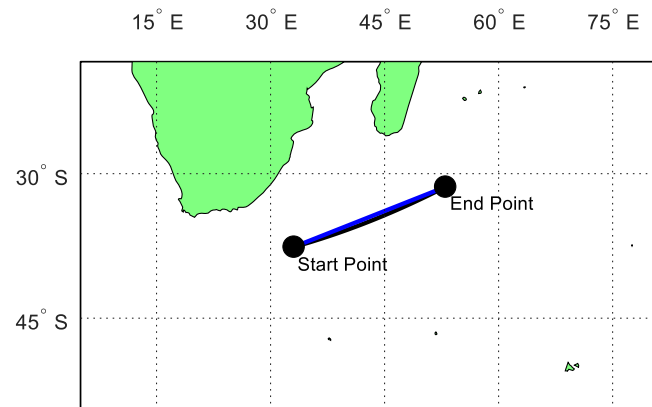


Fig. 1. Reference Route

The availability of daily operational weather forecast data is a critical aspect for assessing the reference scenario required in the adaptive weather routing algorithm. Presently, the community generally adheres to the standardization established by the World Meteorological Organization (WMO), providing all information in a GRIB (GRidded Binary) format, which greatly simplifies the reading of input data.

The model inputs introduced in the route optimization procedure include significant wave height, mean wave period,

and direction of both wind-generated wave components and swell waves.

In the analysis, GRIB files were downloaded for the period from December 7th to December 14th, 2023, with a grid spacing of 0.25°×0.25°, a forecast interval of 3 hours, and an observation period of 7 days. To obtain the GRIB files, the free software XyGrib was utilized, capable of downloading and displaying meteorological data.

Figure 2 is an example of the images depicting changes in environmental conditions in the selected area. These images represent a series of GRIB files showing the temporal trend of significant wave height.

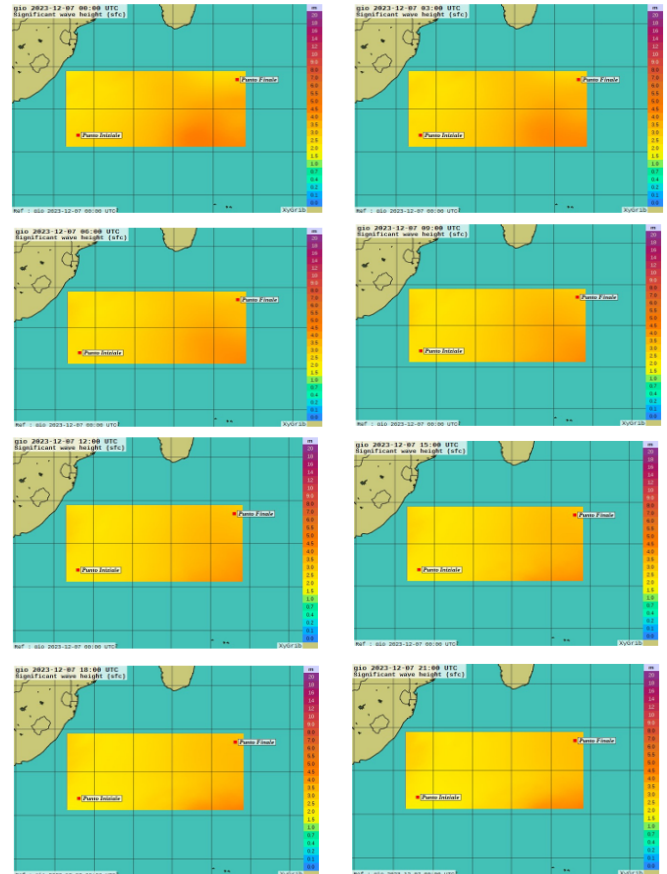


Fig. 2. GRIB Files

V. EXAMPLE OF APPLICATION

Once the input data was established, the route optimization algorithm commenced, assessing a speed of 14.5 knots, correlating to a Froude number of 0.18. Both genetic and Dijkstra optimization algorithms were utilized to compare their distinct characteristics and effectiveness in managing the dynamic complexities inherent in navigating adverse marine conditions. In both cases, the great circle route is highlighted in black, the optimal route is represented in blue, while the black circles represent the departure and arrival points. The genetic algorithm produced the results shown in the figure 3, while the results generated by Dijkstra's algorithm are presented in the figure 4:

Once the optimal routes have been, a percentage index was calculated in order to estimate the improvement in seakeeping criteria in the comparison between the great circle route and the optimized route. The table 3 displays the results generated by both algorithms.

From the comparison between the values assumed by the Seakeeping Performance Index (SPI) in the great circle route and those obtained through optimization with the genetic algorithm, a significant improvement in performance emerged, exceeding 80%. By analyzing the specific improvement index for each criterion, it was possible to note that genetic optimization led to significant progress in the management of slamming and in the prevention of water on deck, representing an important step towards increasing navigation safety in adverse sea conditions.

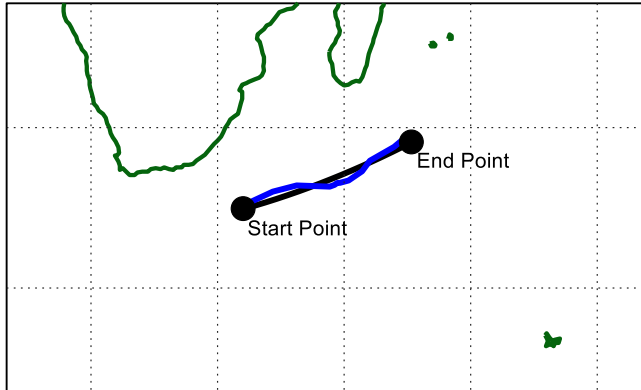


Fig. 3. Genetic algorithm optimal route

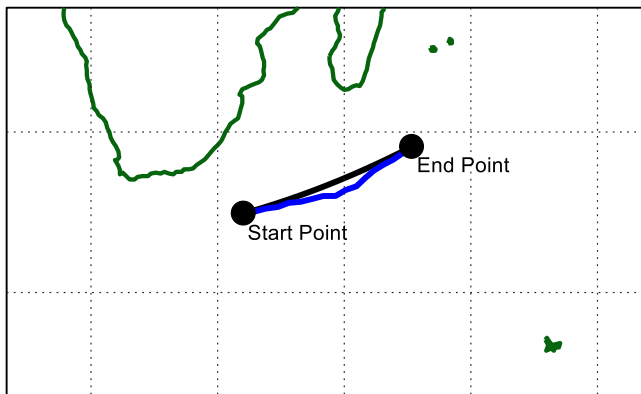


Fig. 4. Dijkstra algorithm optimal route

TABLE 3: PERCENTAGE IMPROVEMENT INDEX

Criteria	Genetic (%)	Dijkstra (%)
RMS pitch	1.9	0.5
RMS roll	8.8	19.1
RMS vertical acceleration at fp	54.8	37.4
Slamming probability	69.7	4.2
Probability of green water on deck	77.7	4.0
MSI	27.2	38.1
Increased distance	8.8	1.8
Voyage duration variation	3.0	-0.05
SPI	87.2	42.6

In accordance with expectations, the genetic optimization resulted in an increase in the distance travelled by the ship, compared to the great circle route, of approximately 93 nautical miles, with a percentage increase of 8%.

Looking at the optimization conducted using the Dijkstra algorithm, however, it is highlighted that by choosing the optimized trajectory rather than the great circle, it is possible to obtain an overall improvement of more than 40%.

Looking at the specific improvement index for each criterion, notable progress emerges in the management of vertical accelerations and passenger comfort, with a direct impact on navigation safety.

In addition to what has been said, the analysis of the route optimized using the Dijkstra algorithm revealed an increase in the distance travelled of 18.7 miles compared to the great circle route, with a percentage increase of 1.8%.

The travel time for the entire route following the great circle, while maintaining a constant speed, is about 73 hours. Assuming the ship operates at a constant engine regime, there is an involuntary speed reduction due to added resistance, leading to an increase in the time needed to reach the final point even along the great circle. In the optimized route, we observe a 3% increase in voyage duration with the genetic algorithm and almost none with the Dijkstra algorithm, as the increase in the route is offset by a smaller speed reduction, resulting from a lesser increase in added resistance.

VI. CONCLUSION

The detailed analysis of the results obtained through the application of Dijkstra and genetic algorithms in the context of weather routing has provided interesting insights into the performance and effectiveness of such approaches. Dijkstra's algorithm has been shown to excel at optimizing routes through a predefined grid of points, allowing rapid identification of optimal solutions. However, its limitation in adapting to dynamic and complex sea conditions emerged during comparison with the genetic algorithm. The latter, not having the restriction of a predefined grid, took around 40 minutes to produce a solution, but showed a notable advantage in optimizing the seakeeping criteria. An important aspect is the notable percentage increase in the seakeeping criteria obtained with the genetic algorithm, especially for the probability of slamming and probability of water on deck. This result highlights the genetic algorithm's ability to adapt to complex variations in sea conditions, providing solutions that significantly improve navigation safety. The differences observed in the outcomes generated by the two algorithms primarily stem from the inclusion, as outlined, of a speed reduction factor induced by varying sea states. Given distinct speed along the two routes and divergent encounter angles, meeting frequency values are consequently altered, directly influencing ship motions. This variance significantly hinges on the apparent frequency of wave encounters experienced by the vessel in motion. The comparison between the execution times of the algorithms highlights a trade-off between time efficiency and quality of the solutions. While Dijkstra shines in terms of speed thanks to its predefined grid, the genetic algorithm takes longer but offers more flexible and adaptable solutions. This discrepancy highlights the importance of carefully considering specific project needs and application conditions when selecting an optimization algorithm.

In summary, the proposed model and methodological approach serve as effective tools for ensuring the safety of navigation for autonomous ships. However, the choice of algorithm to utilize should be carefully assessed, considering the specific project requirements and balancing the trade-offs between processing speed and optimization of security criteria. Certainly, it is of fundamental importance, especially for ships without a crew, to be able to avoid dangerous routes and adapt them to make them safer in adverse sea conditions.

Future developments will include the possibility to implement all seakeeping indices and to select which ones to optimize based on the needs and the service the ship must perform. This will involve incorporating indices that account for nonlinear effects for which there are not yet standardized indices, such as parametric rolling, which is particularly critical for container ships, as in the case of the example ship discussed in the paper. Additionally, further developments will involve the development and implementation of a simulation in the time domain.

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